

FOUNDATIONAL LIMITS

The Problem With Unrestricted Truth

Why classical formal systems cannot contain their own semantic predicates without collapse.

The Dilemma

Every formal language rich enough to encode arithmetic must choose: sacrifice classical logic, or accept semantic incompleteness.

1. The Tarskian Trap

Naive truth requires two intuitive operations:

- **Capture:** If φ holds, then “ φ is true” holds.
- **Release:** If “ φ is true” holds, then φ holds.

The Inevitable Paradox

Pairing capture and release with standard arithmetic yields sentences that assert their own falsehood ($\lambda \iff \neg T(\ulcorner \lambda \urcorner)$).

2. Inconsistency Proof

By the Gödel-Carnap Diagonal Lemma, we derive:

$$\Sigma \vdash \lambda \iff \neg T(\ulcorner \lambda \urcorner)$$

Applying release to λ yields:

$$\Sigma \vdash \lambda \rightarrow \neg \lambda$$

Which reduces instantaneously to:

$$\Sigma \vdash \perp$$

3. How to Break the Loop

Modern formal theories diverge into two architectures:

A. Typed Hierarchies (Tarski)

Stratify truth into metalanguages: $T_0, T_1, T_2, \dots$. Prevents self-reference at the expense of unified expression.

B. Axiomatic Truth (Kripke-Feferman)

Retain a single predicate $T(x)$, but govern it via partial valuation or restricted syntactic reflection.

Architectural Trade-offs

Approach	Primary Cost
Stratified Types	Expressive fragmentation
Fixed Points	Non-classical logic
Axiomatic Systems	Proof-theoretic complexity

Key Takeaway

Expressive power and deductive stability exist in permanent mathematical tension.

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